Relaxation-Driven Cyclic Cosmology (RDCC)

Flagship Paper

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Executive Summary

Relaxation-Driven Cyclic Cosmology (RDCC) is a minimal and predictive framework for a CPT-symmetric Universe. Its construction is based on three structural ingredients:

(a) a bipartite Hilbert-space factorization $\mathcal{H} = \mathcal{H}_+ \otimes \mathcal{H}_-$, relating the visible sector to its CPT-conjugate mirror sector,

(b) a Planck-suppressed inter-sector operator that generates a universal relaxation parameter $\alpha \sim 10^{-2}$,

(c) a cuscuton-induced, ghost-free nonsingular bounce connecting the contracting and expanding phases within the regime of effective field theory.

These ingredients yield a cosmology that is globally CPT-symmetric while reproducing the observed sectoral arrow of time, the dark-matter abundance, the scalar spectral tilt, and percent-level deviations from ΛCDM.

Global CPT Structure and the Arrow of Time. The full quantum state of the Universe is postulated to be a CPT eigenstate. This implies a bipartite Hilbert-space structure in which each sector exhibits its own thermodynamic arrow of time, while the combined system satisfies $S_{\rm tot} = S_+ + S_- = \text{const}$. The bounce corresponds to the hypersurface of minimal coarse-grained entropy. The relaxation parameter α quantifies the rate of entanglement exchange between the sectors and flows to zero at the bounce, restoring maximal correlation and manifest CPT symmetry.

Effective Field Theory and the Origin of α. The inter-sector coupling is generated by the unique dimension-six operator $\mathcal{L}_{\rm int} = M^{-2} T^{(+)}_{\mu\nu} T^{(-)\mu\nu}$, arising from integrating out heavy gravitationally coupled fields. Its renormalization-group flow exhibits an infrared fixed point $\alpha(\mu) \to \alpha_* \sim 10^{-2}$, providing a universal explanation for the magnitude of the relaxation parameter. Microscopically, α admits complementary interpretations: (i) loop-suppressed coefficient, (ii) entanglement-leakage rate, and (iii) RG-attractor controlling sectoral equilibration.

Gravity as the Global Bridge. Gravity is the only interaction that couples to the full CPT-symmetric state. The global Einstein equations involve the sum of the two stress-energy tensors, $G_{\mu\nu} = 8\pi G_N\,(T^{(+)}_{\mu\nu} + T^{(-)}_{\mu\nu})$, while a visible-sector observer perceives an effective Newton constant $G_{\rm eff} = G_N/(1+\rho_-/\rho_+)$. The mirror sector therefore appears as a gravitational shadow, providing a structural explanation for dark matter without introducing new particle species.

Bounce Dynamics and Phase-Space Stability. A cuscuton-like operator generates controlled NEC violation without additional propagating degrees of freedom. The modified Friedmann equation,

$$H^2 = \frac{\rho_{\rm tot}}{3M_{\rm Pl}^2}\left(1 - \frac{\rho_{\rm tot}}{\rho_{\rm crit}}\right),$$

ensures a smooth turnaround at $\rho_{\rm tot} = \rho_{\rm crit}$. A phase-space analysis shows that the CPT-symmetric trajectory is a dynamical attractor: antisymmetric perturbations between the sectors are damped by the relaxation dynamics, and the bounce is stable for a wide range of initial data.

Perturbations and Gravitational Waves. RDCC predicts a nearly scale-invariant scalar spectrum with $n_s - 1 = -2\alpha$, and a blue-tilted stochastic gravitational-wave background peaked in the millihertz range. A distinctive signature is the presence of log-periodic oscillations in the tensor spectrum,

$$\Omega_{\rm GW}(k) = \Omega_0(k)\ [1 + \varepsilon \cos(\omega \ln(k/k_b) + \phi)]\,,$$

arising from interference between the two CPT-related tensor sectors across the bounce.

Phenomenology and Falsifiability. All deviations from ΛCDM are governed by the single parameter α. RDCC predicts correlated signatures in the scalar spectral tilt, dark-radiation excess $\Delta N_{\rm eff}$, the growth-rate observable $f\sigma_8(z)$, the dark-matter to baryon ratio, and the stochastic gravitational-wave background. The framework is sharply falsifiable: any inconsistency among these correlated predictions would rule out RDCC, while detection of log-periodic GW modulations would strongly support it.

Abstract

Fixing the relaxation parameter from the observed scalar tilt yields $\alpha \simeq 0.0175$, in excellent agreement with the expected infrared fixed-point value. This single parameter governs all correlated deviations from ΛCDM in Relaxation-Driven Cyclic Cosmology (RDCC), providing a direct observational anchor for the framework. In the accompanying observational analysis (Companion V), we present a prototype global fit combining representative constraints from the scalar tilt, dark radiation, large-scale structure, and upper bounds on the stochastic gravitational-wave background. The resulting posterior peaks at $\alpha \simeq 0.016$, fully consistent with the analytic estimate and demonstrating compatibility with current data.

RDCC offers a minimal and predictive realization of a CPT-symmetric Universe. The framework is built on three structural ingredients: (i) a bipartite Hilbert-space factorization relating the visible sector to its CPT-conjugate mirror sector, (ii) a Planck-suppressed intersector operator generating the universal relaxation parameter α, and (iii) a cuscuton-induced, ghost-free nonsingular bounce connecting the contracting and expanding phases within the regime of effective field theory.

The global CPT structure implies conservation of the total coarse-grained entropy, while each sector exhibits its own thermodynamic arrow of time. The bounce corresponds to the hypersurface of minimal entropy, where $\alpha \to 0$ and the two sectors become maximally correlated. Gravity acts as the unique global interaction, coupling to the sum of the two stress-energy tensors and giving rise to an effective Newton constant $G_{\text{eff}} = G_N/(1 + \rho_-/\rho_+)$. The mirror sector therefore appears as a gravitational shadow, providing a structural explanation for dark matter without introducing new particle species.

RDCC predicts a nearly scale-invariant scalar spectrum with $n_s - 1 = -2\alpha$, a blue-tilted stochastic gravitational-wave background peaked in the millihertz range, and distinctive log-periodic oscillations in the tensor spectrum arising from interference between the two CPT-related sectors across the bounce. All deviations from ΛCDM follow from the single parameter α, yielding a sharply falsifiable, one-parameter correlation structure across early- and late-time cosmology.

1 Introduction

The standard ΛCDM model provides an exceptionally successful description of the expansion history and the statistics of cosmological perturbations. Nevertheless, several foundational questions remain unresolved within this framework. These include the origin of the thermodynamic arrow of time, the nature of the initial singularity, the Tolman entropy problem, the structural origin of dark matter, and the mechanism underlying the baryon asymmetry. Existing proposals typically address these puzzles through independent extensions of the standard cosmological model—such as inflation, modified gravity, or new particle species— without explaining why these issues arise together or why their proposed solutions remain uncorrelated.

Relaxation-Driven Cyclic Cosmology (RDCC) offers a minimal and internally consistent completion of ΛCDM that addresses these questions within standard General Relativity and effective field theory (EFT). The framework is built on a single structural postulate: the global quantum state of the Universe is a CPT eigenstate. The minimal EFT-level realization of such a state requires a bipartite Hilbert-space factorization,

$$\mathcal{H} = \mathcal{H}_+ \otimes \mathcal{H}_-, \tag{1}$$

where $\mathcal{H}_+$ describes the visible sector and $\mathcal{H}_-$ its CPT-conjugate mirror sector. The two sectors propagate on the same spacetime metric and interact only through gravity and a Planck-suppressed irrelevant operator. This construction introduces no new propagating degrees of freedom; instead, it specifies the global structure of the cosmological state.

A defining feature of RDCC is its one-parameter structure. A single infrared relaxation parameter $\alpha \sim 10^{-2}$, generated by the leading dimension-six inter-sector operator, governs correlated deviations across multiple cosmological epochs. These include the scalar spectral tilt $n_s - 1 = -2\alpha$, the dark-radiation excess $\Delta N_{\rm eff} \sim \alpha$, percent-level corrections to the growth-rate observable $f\sigma_8(z)$, the dark-matter to baryon ratio, and the amplitude and structure of the stochastic gravitational-wave background. The relaxation parameter admits complementary microscopic interpretations—as a loop-suppressed coefficient, as a measure of entanglement leakage between the sectors, and as an infrared fixed point of the renormalization-group flow. These interpretations reinforce the universality of α and explain why a single dimensionless parameter governs correlated deviations from ΛCDM.

A second structural ingredient of RDCC is the cuscuton-induced nonsingular bounce. The cuscuton arises naturally from integrating out heavy gravitationally coupled fields and provides a controlled, ghost-free mechanism for violating the null energy condition. The resulting modified Friedmann equation yields a smooth transition between contraction and expansion at a critical density. A phase-space analysis shows that the CPT-symmetric trajectory is a dynamical attractor: antisymmetric perturbations between the sectors are damped by the relaxation dynamics, and the bounce is stable for a wide range of initial conditions.

The third structural ingredient is the role of gravity as the unique global interaction. While gauge interactions act strictly within a single sector, the metric couples to the sum of the two stress-energy tensors,

$$G_{\mu\nu} = 8\pi G_N \left(T^{(+)}_{\mu\nu} + T^{(-)}_{\mu\nu} \right). \tag{2}$$

A visible-sector observer therefore perceives an effective Newton constant,

$$G_{\rm eff} = \frac{G_N}{1 + \rho_-/\rho_+}, \tag{3}$$

and infers the presence of an additional gravitational component. This gravitational shadow of the mirror sector behaves exactly like cold dark matter, providing a structural explanation for the observed dark-matter abundance without introducing new particle species.

RDCC predicts a nearly scale-invariant scalar spectrum, a blue-tilted stochastic gravitational-wave background peaked in the millihertz range, and distinctive log-periodic oscillations in the tensor spectrum arising from interference between the two CPT-related sectors across the bounce. These oscillations provide a clear observational target for LISA and constitute a smoking-gun signature of the RDCC framework.

Structure of This Paper

Section 2 develops the global CPT structure, the bipartite Hilbert-space factorization, and the thermodynamic interpretation of the arrow of time. Section 3 constructs the effective field theory, including the origin and RG flow of the relaxation parameter and the microscopic derivation of the cuscuton operator. Section 4 develops the global and sectoral Einstein equations, derives the effective Newton constant, and defines the gravitational shadow of the mirror sector. Section 5 analyzes the bounce dynamics and phase-space stability. Section 6 develops the scalar and tensor perturbations, including the analytical origin of log-periodic gravitational-wave oscillations. Section 7 summarizes the phenomenological predictions and falsifiability criteria. Section 8 concludes with a discussion of the structural implications of the framework.

Nomenclature

List of Symbols

Table 1: List of symbols used in this work.

Symbol	Description	First Appearance
$a(t)$	Scale factor	Sec. 5
c_s^2	Effective sound speed of scalar perturbations	Sec. 6
$G_{\mu\nu}$	Einstein tensor	Sec. 4
G_{eff}	Effective Newton constant	Sec. 4
G_N	Newton constant	Sec. 4
H	Hubble parameter	Sec. 5
$\mathcal{H}_\pm$	Visible and mirror Hilbert-space sectors	Sec. 2
k	Comoving wavenumber	Sec. 6
$K(t)$, $G(t)$	Kinetic coefficients in scalar perturbation action	Sec. 6
M	Heavy UV scale generating the inter-sector operator	Sec. 3
M_{Pl}	Reduced Planck mass	Sec. 5
$\Omega_{\text{GW}}(k)$	Stochastic gravitational-wave spectrum	Sec. 6
$\rho_\pm$	Energy densities of visible and mirror sectors	Sec. 4
ρ_{crit}	Critical density at which the bounce occurs	Sec. 5
ρ_{tot}	Total energy density of both sectors	Sec. 5
$S_\pm$	Coarse-grained entropies of the two sectors	Sec. 2
S_{tot}	Total coarse-grained entropy	Sec. 2
$T_{\mu\nu}^{(\pm)}$	Stress-energy tensors of visible and mirror sectors	Sec. 4
$t_\pm$	Sectoral time coordinates	Sec. 2
$V_{\text{eff}}(\phi)$	Radiatively generated scalar potential	Sec. 3
α	Infrared relaxation parameter	Sec. 3
ε	Amplitude of log-periodic tensor modulation	Sec. 6
ϕ	Effective scalar field driving background dynamics	Sec. 3
ζ	Curvature perturbation	Sec. 6

List of Acronyms

Table 2: List of acronyms used in this work.

Acronym	Description	First Appearance
CMB	Cosmic Microwave Background	Sec. 6
CPT	Charge-Parity-Time symmetry	Sec. 2
EFT	Effective Field Theory	Sec. 3
FLRW	Friedmann–Lemaître–Robertson–Walker metric	Sec. 4
LISA	Laser Interferometer Space Antenna	Sec. 6
RDCC	Relaxation-Driven Cyclic Cosmology	Title
RG	Renormalization Group	Sec. 3
SGWB	Stochastic Gravitational-Wave Background	Sec. 6

2 Foundations of the RDCC Framework

Relaxation-Driven Cyclic Cosmology (RDCC) is built on a single structural postulate: the full quantum state of the Universe is a CPT eigenstate. This assumption determines the global architecture of the cosmological state and implies a bipartite Hilbert-space factorization, sectoral thermodynamic arrows of time, and a universal relaxation dynamics governed by the parameter α. In this section we develop the foundational ingredients of the framework.

2.1 Global CPT Postulate

The central structural assumption of RDCC is that the global quantum state $|\Psi\rangle$ satisfies

$$\mathrm{CPT}\,|\Psi\rangle = |\Psi\rangle. \tag{4}$$

In a cosmological setting this implies that the global state must contain both a sector and its CPT-conjugate counterpart. The minimal EFT-level realization of such a state requires a bipartite Hilbert-space factorization,

$$\mathcal{H} = \mathcal{H}_+ \otimes \mathcal{H}_-, \tag{5}$$

where $\mathcal{H}_+$ describes the visible sector and $\mathcal{H}_-$ its CPT-conjugate mirror sector.

Geometric Neutrality of the Time Coordinate. The global spacetime geometry in RDCC is a time-direction-neutral continuum. The fundamental metric does not distinguish between the two orientations of the time axis. Only after sectoral projection, induced by the asymmetric coupling of the global scalar field, do local observers in sector A experience a positive thermodynamic time direction (t_+), while observers in the CPT-conjugate sector B experience the opposite orientation (t_-). The global spacetime remains the neutral geometric carrier of both sectoral dynamics, analogous to the spatial coordinates (x, y, z), which become $(x_\pm)$ only after projection.

A generic CPT-symmetric state may be written in Schmidt-like form,

$$|\Psi\rangle = \sum_i c_i\, |i\rangle_+ \otimes |i\rangle_-, \tag{6}$$

where the coefficients c_i encode the degree of sectoral entanglement. In the infrared limit of the relaxation dynamics, $\alpha \to 0$, the two sectors become maximally correlated and the global state approaches a pure CPT eigenstate.

For clarity we distinguish three conceptual layers of the framework: (i) the global CPT postulate, (ii) the bipartite Hilbert-space representation, (iii) the physical interpretation in terms of entropy flow, thermodynamic arrows of time, and the gravitational shadow. These layers are logically distinct.

2.2 Sectoral Observers and Reduced Density Matrices

Observers exist only within a single sector. A visible-sector observer has access only to operators acting on $\mathcal{H}_+$ and therefore perceives the reduced density matrix

$$\rho_+ = \mathrm{Tr}_-(|\Psi\rangle\langle\Psi|)\,. \tag{7}$$

Gauge interactions act strictly within a single sector, while gravity couples to the sum of the two stress-energy tensors. This structural asymmetry is the origin of the gravitational shadow of the mirror sector and underlies the interpretation of dark matter in RDCC.

The bipartite factorization is not enforced by CPT symmetry alone; rather, it is the minimal structural realization that allows for a well-defined entropy minimum and sectoral thermodynamic arrows of time.

2.3 Thermodynamic Interpretation and the Arrow of Time

The thermodynamic arrow of time is a sectoral concept rather than a fundamental asymmetry of the microscopic laws. The global CPT-symmetric state satisfies

$$S_{\text{tot}} = S_+ + S_- = \text{const}, \tag{8}$$

where S_+ and S_- denote the coarse-grained entropies of the visible and mirror sectors. Each sector experiences an increase of entropy along its own time orientation,

$$\frac{dS_+}{dt_+} > 0, \qquad \frac{dS_-}{dt_-} > 0, \tag{9}$$

while the combined system remains globally time-symmetric.

The bounce corresponds to the hypersurface of minimal coarse-grained entropy. At this point the two sectors become maximally correlated and the relaxation parameter approaches its infrared value,

$$\alpha(t_{\text{b}}) \to 0. \tag{10}$$

This entropic interpretation provides a natural origin for the sectoral arrows of time without invoking special initial conditions.

2.4 The Relaxation Parameter as Entanglement Leakage

The relaxation parameter α quantifies the rate of energy, entropy, and information exchange between the two sectors. In the bipartite description one may view α as a dimensionless measure of entanglement leakage,

$$\alpha \sim \frac{\Delta S}{S_{\text{tot}}}, \tag{11}$$

where ΔS denotes the entropy transferred between sectors over a Hubble time. In the infrared limit $\alpha \to 0$, the two sectors become maximally correlated and the global state approaches a pure CPT eigenstate.

This interpretation links the relaxation dynamics to the global structure of the quantum state and clarifies why the bounce acts as a structural boundary: it is the moment at which the coarse-grained entropy of the combined system attains its minimum.

2.5 Semi-Classical Path-Integral Motivation

Although RDCC adopts the CPT-eigenstate condition as a structural postulate, it admits a heuristic motivation from a semi-classical path-integral perspective. In a Wheeler–DeWitt or Hartle–Hawking-type formulation, the cosmological wave functional may be written schematically as

$$\Psi[g,\phi] = \int \mathcal{D}g\,\mathcal{D}\phi\; e^{\,iS[g,\phi]}. \tag{12}$$

Configurations satisfying

$$S[g,\phi] = S[\text{CPT}(g,\phi)] \tag{13}$$

represent stationary contributions to the path integral and naturally include time-symmetric bounce geometries. This argument does not constitute a derivation of the CPT-eigenstate structure but provides a structural motivation for adopting it at the EFT level.

3 RDCC and General Relativity: Structural Compatibility and Temporal Neutrality

The Relaxation-Driven Cyclic Cosmology (RDCC) is fully compatible with the geometric structure of General Relativity (GR). Unlike many alternative cosmological frameworks, RDCC does not modify Einstein's field equations, introduce additional geometric degrees of freedom, or alter the causal structure of spacetime. Instead, RDCC extends GR by providing a microscopic interpretation of the global state of the Universe and the emergence of thermodynamic time.

At the geometric level, RDCC assumes the validity of the Einstein equations

$$G_{\mu\nu} = 8\pi G\, T_{\mu\nu},$$

with the stress-energy tensor receiving contributions from two CPT-conjugate sectors. The bipartite structure of the global state is defined on the tensor product Hilbert space

$$\mathcal{H}_{\text{global}} = \mathcal{H}_+ \otimes \mathcal{H}_-$$

but both sectors propagate on the same spacetime geometry. The metric structure remains single, classical, and fully governed by GR.

The key extension introduced by RDCC concerns the temporal interpretation of the global state. Before sectoral projection, the time coordinate is geometrically neutral: it carries no thermodynamic arrow and no preferred direction of entropy increase. This neutrality is consistent with the conformal structure of GR, in which null directions define the causal boundary of spacetime.

RDCC identifies the speed of light c as the boundary of temporal neutrality. At the null limit, the thermodynamic arrow is not yet defined, because the sectoral projection has not taken place. Only for subluminal observers does the projection onto one of the two CPT-conjugate sectors occur, breaking the conformal symmetry and producing a definite thermodynamic time direction. In this sense, the emergence of time in RDCC is a thermodynamic refinement of the causal structure already present in GR.

The relaxation dynamics provide the mechanism by which the bipartite global state decoheres into two classical sectors with opposite thermodynamic arrows. This process does not alter the Einstein equations; it determines which sectoral stress-energy tensor contributes to the classical geometry experienced by a given observer. The geometry remains classical and single-valued, while the thermodynamic arrow becomes sector-specific.

RDCC therefore complements GR rather than modifying it. GR governs the geometric evolution of spacetime, while RDCC explains the emergence of thermodynamic time, the sectoral structure of matter, and the origin of cosmic asymmetries. The two frameworks are structurally compatible: GR provides the geometric backbone, and RDCC provides the microscopic interpretation of the global state and its sectoral projections.

A further structural aspect concerns the role of gravity in the emergence of sectoral time. Since the metric couples to the sum of the two stress-energy tensors, gravity provides the unique global coordinate structure shared by both sectors. At the null limit, where $v = c$, the time coordinate remains geometrically neutral and no sectoral projection has taken place. Only for subluminal observers does the projection onto one of the two CPT-conjugate sectors occur, producing effective mass and a definite thermodynamic arrow of time. In this sense, gravity anchors the sectoral projection within the global spacetime geometry: once mass is generated ($v < c$), the metric fixes the sectoral contribution to the Einstein equations and stabilizes the emergent time orientation. This refinement links the temporal neutrality at c to the geometric universality of gravity and clarifies why the thermodynamic arrow is a sectoral, rather than geometric, property.

4 Effective Field Theory Structure

The effective field theory (EFT) underlying RDCC is deliberately minimal. It introduces no new propagating degrees of freedom beyond those of the Standard Model and General Relativity. Instead, the framework relies on the structural consequences of the global CPT postulate, the bipartite Hilbert-space factorization, and a single Planck-suppressed inter-sector operator. These ingredients generate a universal relaxation parameter $\alpha \sim 10^{-2}$, a cuscuton-induced nonsingular bounce, and correlated deviations from ΛCDM across early- and late-time cosmology.

4.1 Inter-Sector Operator and the Origin of α

At leading order in the EFT expansion, the interaction between the visible and mirror sectors is described by the unique dimension-six operator consistent with general covariance and CPT symmetry,

$$\mathcal{L}_{\text{int}} = \frac{1}{M^2} T^{(+)}_{\mu\nu} T^{(-)\mu\nu}, \tag{14}$$

where M denotes a heavy mass scale associated with ultraviolet degrees of freedom integrated out at low energies. This operator induces a slow exchange of energy, entropy, and information between the two sectors. Its macroscopic effect is captured by a single dimensionless relaxation parameter α, which controls the rate at which the two sectors approach dynamical and thermal equilibrium.

The relaxation parameter is not a free input. Instead, it is generated radiatively and exhibits a simple renormalization-group structure. At one-loop order, the natural size of the coefficient of the operator is

$$\alpha \sim \frac{1}{16\pi^2}, \tag{15}$$

which lies in the range required for RDCC phenomenology.

For clarity we distinguish three conceptual levels associated with the operator: (i) the EFT level, where it acts as a low-energy interaction; (ii) the RG level, where its coefficient flows toward an infrared fixed point; (iii) the UV level, where heavy fields generate the operator radiatively. These layers are logically distinct even though they combine into a single effective description.

4.2 Microscopic Interpretations of the Relaxation Parameter

The relaxation parameter α admits three complementary microscopic interpretations:

(i) Loop-suppressed coefficient. Integrating out heavy gravitationally coupled fields generates the operator with a loop-suppressed coefficient. This explains why α is naturally small and universal.

(ii) Entanglement leakage. Within the bipartite Hilbert-space structure, α quantifies the rate of entanglement leakage between the two sectors,

$$\alpha \sim \frac{\Delta S}{S_{\text{tot}}}, \tag{16}$$

where ΔS denotes the entropy transferred between sectors over a Hubble time. This interpretation links the relaxation dynamics to the global structure of the quantum state.

(iii) Infrared fixed point of the RG flow. The effective coupling satisfies the renormalization-group equation

$$\frac{d\alpha}{d\ln\mu} = -c\,\alpha^2 + \mathcal{O}(\alpha^3), \qquad c > 0, \tag{17}$$

which implies the existence of an infrared fixed point,

$$\alpha(\mu) \longrightarrow \alpha_* \sim 10^{-2}. \tag{18}$$

This RG-attractor explains the universality of α across cosmological epochs and its insensitivity to ultraviolet details.

Throughout this work, $\alpha(H)$ denotes a scale-dependent relaxation parameter: it flows to $\alpha \to 0$ near the bounce ($H \to 0$) and approaches its infrared fixed point α_* at late times.

4.3 Scalar Sector and Effective Lagrangian

The cosmological dynamics are driven by a single real scalar degree of freedom ϕ, interpreted as an effective Higgs-like radial mode. The EFT Lagrangian takes the form

$$\mathcal{L}_\phi = -\frac{1}{2}(\partial\phi)^2 - V_{\text{eff}}(\phi) + \mathcal{L}_{\text{cuscuton}}, \tag{19}$$

where V_{eff} is a radiatively generated potential and $\mathcal{L}_{\text{cuscuton}}$ denotes the cuscuton-like operator responsible for the nonsingular bounce.

The effective potential may be written schematically as

$$V_{\text{eff}}(\phi) = \lambda(\phi^2 - v^2)^2 + \epsilon\,\phi + \beta\, e^{-\gamma\phi} + V_{\text{rad}}(\phi), \tag{20}$$

where the quartic term stabilizes the potential, the linear and exponential terms encode radiative corrections, and V_{rad} denotes additional loop-induced contributions. The precise form of the potential is not essential for the structural arguments of RDCC; only its smoothness and radiative origin are required.

The cuscuton term provides a controlled, ghost-free mechanism for violating the null energy condition and ensures that the bounce occurs within the regime of validity of effective field theory.

5 Gravity as the Global Bridge

Gravity plays a unique structural role in RDCC: it is the only interaction that couples to the full CPT-symmetric state. Gauge interactions act strictly within a single sector, while the metric couples to the sum of the two stress-energy tensors. This asymmetry is responsible for the gravitational shadow of the mirror sector and provides a structural explanation for dark matter without introducing new particle species.

The global Einstein equations take the form

$$G_{\mu\nu} = 8\pi G_N \left(T^{(+)}_{\mu\nu} + T^{(-)}_{\mu\nu}\right), \tag{21}$$

where $T^{(+)}_{\mu\nu}$ and $T^{(-)}_{\mu\nu}$ denote the stress-energy tensors of the visible and mirror sectors. A visible-sector observer, who has access only to operators acting on $\mathcal{H}_+$, perceives an effective Newton constant

$$G_{\text{eff}} = \frac{G_N}{1 + \rho_-/\rho_+}, \tag{22}$$

where ρ_+ and ρ_- denote the energy densities of the two sectors. The mirror sector therefore appears as an additional gravitational component, behaving exactly like cold dark matter.

5.1 Global and Sectoral Einstein Equations

The global metric $g_{\mu\nu}$ is shared by both sectors. The Einstein equations governing the full CPT-symmetric state are

$$G_{\mu\nu}[g] = 8\pi G_N \left(T^{(+)}_{\mu\nu}[g] + T^{(-)}_{\mu\nu}[g]\right). \tag{23}$$

A sectoral observer, however, reconstructs the gravitational dynamics from the reduced density matrix ρ_+ or ρ_-. For a visible-sector observer, the effective Einstein equations take the form

$$G_{\mu\nu}[g] = 8\pi G_{\text{eff}}\, T^{(+)}_{\mu\nu}[g], \tag{24}$$

with G_{eff} given by the ratio of sectoral energy densities. This structural reinterpretation replaces the need for new dark-matter particles by a purely geometric effect arising from the bipartite Hilbert-space structure.

5.2 Gravitational Shadow of the Mirror Sector

The mirror sector contributes to the global stress-energy tensor but remains invisible to gauge interactions in the visible sector. Its gravitational effect is therefore perceived as a shadow component. The ratio

$$\frac{\rho_-}{\rho_+} \tag{25}$$

determines the magnitude of this shadow and sets the effective dark-matter abundance. Since both sectors share the same metric, the gravitational shadow behaves like cold dark matter at the background and perturbation levels.

This interpretation is structural rather than dynamical: no new fields or interactions are introduced. The dark-matter abundance arises from the global CPT-symmetric state and the bipartite Hilbert-space factorization.

5.3 Consistency with the Relaxation Dynamics

The relaxation parameter α governs the rate of energy and entropy exchange between the sectors. As $\alpha \to 0$ near the bounce, the two sectors become maximally correlated and the gravitational shadow becomes symmetric. At late times, $\alpha \to \alpha_*$, and the ratio ρ_-/ρ_+ stabilizes at the value required to reproduce the observed dark-matter abundance.

Thus the gravitational shadow, the effective Newton constant, and the sectoral interpretation of dark matter are direct consequences of the global CPT structure and the relaxation dynamics.

6 Bounce Dynamics and Phase-Space Stability

The nonsingular bounce is a structural component of RDCC. It arises from a cuscuton-like operator that provides controlled violation of the null energy condition (NEC) without introducing additional propagating degrees of freedom. The bounce connects the contracting and expanding phases within the regime of validity of effective field theory and ensures that the global CPT-symmetric trajectory is dynamically stable.

6.1 Modified Friedmann Equation

The effective background dynamics are governed by the modified Friedmann equation

$$H^2 = \frac{\rho_{\text{tot}}}{3M_{\text{Pl}}^2}\left(1 - \frac{\rho_{\text{tot}}}{\rho_{\text{crit}}}\right), \tag{26}$$

where $\rho_{\text{tot}} = \rho_+ + \rho_-$ denotes the combined energy density of the two sectors and ρ_{crit} is the critical density at which the bounce occurs. The factor in parentheses ensures a smooth turnaround at $\rho_{\text{tot}} = \rho_{\text{crit}}$, with $H = 0$ and $\dot{H} > 0$, signaling a transition from contraction to expansion.

The bounce is nonsingular: the scale factor $a(t)$, the Hubble parameter $H(t)$, and all curvature invariants remain finite throughout the evolution. This behavior is a direct consequence of the cuscuton term, which modifies the Hamiltonian constraint without introducing new dynamical modes.

6.2 Cuscuton-Induced NEC Violation

The cuscuton operator provides a controlled mechanism for violating the null energy condition. Its defining property is the absence of a propagating kinetic term, which ensures that no ghost or gradient instabilities are introduced. Instead, the cuscuton modifies the constraint equations in such a way that the bounce occurs at a finite and calculable energy density.

The NEC-violating contribution is purely geometric and does not rely on exotic matter fields. This allows the bounce to occur within the domain of validity of effective field theory and avoids the pathologies typically associated with NEC-violating models.

6.3 Phase-Space Structure and Stability

The dynamical system describing the background evolution of the two sectors may be written schematically as

$$\dot{\mathbf{X}} = \mathbf{F}(\mathbf{X}), \tag{27}$$

where $\mathbf{X}$ denotes the set of background variables and $\mathbf{F}$ encodes the coupled evolution equations. Linearizing around the CPT-symmetric trajectory yields

$$\delta\dot{\mathbf{X}} = \mathbf{M}\,\delta\mathbf{X}, \tag{28}$$

where $\mathbf{M}$ is the stability matrix. The relaxation dynamics ensure that antisymmetric perturbations between the sectors are damped,

$$\mathrm{Re}\,\lambda_i < 0, \tag{29}$$

for all eigenvalues λ_i of $\mathbf{M}$. Thus the CPT-symmetric trajectory acts as a dynamical attractor.

This stability is a direct consequence of the relaxation parameter $\alpha(H)$, which flows to zero near the bounce. As $\alpha \to 0$, the two sectors become maximally correlated, suppressing antisymmetric deviations and stabilizing the bounce.

6.4 Consistency with the Global CPT Structure

The bounce corresponds to the hypersurface of minimal coarse-grained entropy, where the two sectors become maximally correlated and the global state approaches a pure CPT eigenstate. The relaxation parameter satisfies

$$\alpha(t_{\mathrm{b}}) \to 0, \tag{30}$$

ensuring that the entanglement structure is symmetric across the bounce.

The combination of the cuscuton-induced NEC violation, the modified Friedmann equation, and the relaxation dynamics yields a nonsingular, stable, and structurally consistent bounce. This mechanism replaces the initial singularity of standard cosmology with a CPT-symmetric transition that preserves the global architecture of RDCC.

7 Perturbations and Gravitational Waves

The perturbation sector of RDCC inherits its structure from the bipartite Hilbert-space representation, the global CPT symmetry, and the relaxation dynamics. Scalar and tensor modes propagate on the shared FLRW background, while their amplitudes and correlations are influenced by the inter-sector coupling and the bounce. The resulting predictions include a nearly scale-invariant scalar spectrum, a blue-tilted stochastic gravitational-wave background, and log-periodic oscillations arising from interference between the two CPT-related tensor sectors.

7.1 Scalar Perturbations

Scalar perturbations are described by the curvature perturbation ζ. The quadratic action takes the form

$$S_\zeta = \frac{1}{2}\int dt\, d^3x\, a^3 \left[K(t)\,\dot{\zeta}^2 - G(t)\,\frac{(\nabla\zeta)^2}{a^2}\right], \tag{31}$$

where $K(t)$ and $G(t)$ are background-dependent kinetic coefficients. The effective sound speed is given by

$$c_s^2 = \frac{G(t)}{K(t)}. \tag{32}$$

The scalar spectral index is determined by the relaxation parameter,

$$n_s - 1 = -2\alpha, \tag{33}$$

yielding a nearly scale-invariant spectrum for $\alpha \sim 10^{-2}$. This relation is a structural prediction of RDCC and reflects the universal role of the relaxation dynamics across cosmological epochs.

7.2 Tensor Perturbations

Tensor perturbations h_{ij} propagate freely in each sector but acquire sectoral correlations through the global CPT structure. The quadratic action is

$$S_h = \frac{M_{\rm Pl}^2}{8}\int dt\, d^3x\, a^3 \left[\dot{h}_{ij}^2 - \frac{(\nabla h_{ij})^2}{a^2}\right]. \tag{34}$$

The tensor spectrum is blue-tilted due to the contracting phase preceding the bounce. The amplitude and tilt are determined by the background evolution and the relaxation dynamics, with the bounce imprinting characteristic features on the spectrum.

7.3 Log-Periodic Tensor Modulation

A distinctive prediction of RDCC is the presence of log-periodic oscillations in the tensor spectrum. These arise from interference between the two CPT-related tensor sectors across the bounce. The resulting spectrum takes the form

$$\Omega_{\rm GW}(k) = \Omega_0(k)\,\left[1 + \varepsilon\cos\!\left(\omega \ln\frac{k}{k_b} + \phi\right)\right], \tag{35}$$

where ε is the modulation amplitude, ω the logarithmic frequency, k_b the characteristic bounce scale, and ϕ a phase shift.

These oscillations are a direct consequence of the global CPT structure and the bipartite entanglement across the bounce. They provide a clear observational target for LISA and constitute a smoking-gun signature of the RDCC framework.

7.4 Stochastic Gravitational-Wave Background

The stochastic gravitational-wave background (SGWB) predicted by RDCC is blue-tilted and peaks in the millihertz range. The peak frequency is determined by the bounce scale,

$$f_{\rm peak} \sim \frac{k_b}{2\pi a_0}, \tag{36}$$

placing it squarely within the sensitivity band of LISA.

The combination of a blue tilt, a characteristic peak, and log-periodic modulations yields a highly distinctive SGWB signature. Detection of any of these features would provide strong evidence for the RDCC framework.

7.5 Consistency with the Bounce and Relaxation Dynamics

The perturbation sector is fully consistent with the bounce dynamics and the relaxation structure. Near the bounce, the relaxation parameter satisfies

$$\alpha(t_{\mathrm{b}}) \to 0, \tag{37}$$

ensuring maximal correlation between the two sectors and enforcing the symmetry conditions required for the log-periodic tensor modulation.

At late times, $\alpha \to \alpha_*$, stabilizing the scalar spectral tilt and the amplitude of the SGWB. The perturbation sector therefore provides a direct observational window into the relaxation dynamics and the global CPT structure of RDCC.

8 Phenomenology and Falsifiability

All observable deviations from ΛCDM in RDCC are governed by the single relaxation parameter α. Its universal role links early- and late-time cosmology, producing a tightly correlated prediction structure across multiple observables. This correlation pattern is a defining feature of the framework and renders RDCC sharply falsifiable.

8.1 Correlated Predictions

The relaxation parameter determines the scalar spectral tilt,

$$n_s - 1 = -2\alpha, \tag{38}$$

yielding $n_s \simeq 0.965$ for $\alpha \sim 0.017$. The same parameter governs the dark-radiation excess,

$$\Delta N_{\mathrm{eff}} \sim \alpha, \tag{39}$$

and induces percent-level corrections to the growth-rate observable $f\sigma_8(z)$.

The dark-matter to baryon ratio is fixed by the ratio of sectoral energy densities,

$$\frac{\Omega_{\mathrm{DM}}}{\Omega_b} = \frac{\rho_-}{\rho_+}, \tag{40}$$

which stabilizes dynamically as $\alpha(H)$ approaches its infrared fixed point. This provides a structural explanation for the observed dark-matter abundance without introducing new particle species.

The stochastic gravitational-wave background (SGWB) inherits its structure from the bounce and the global CPT symmetry. RDCC predicts a blue-tilted spectrum peaked in the millihertz range, with log-periodic oscillations of the form

$$\Omega_{\mathrm{GW}}(k) = \Omega_0(k)\,\left[1 + \varepsilon\cos\!\left(\omega \ln\frac{k}{k_b} + \phi\right)\right]. \tag{41}$$

These oscillations arise from interference between the two CPT-related tensor sectors across the bounce and constitute a distinctive observational signature.

8.2 Observational Windows

The correlated prediction structure of RDCC spans multiple observational channels:

- scalar spectral tilt n_s (CMB),
- dark-radiation excess ΔN_{eff} (CMB, BBN),

- growth-rate observable $f\sigma_8(z)$ (LSS),
- dark-matter to baryon ratio (background cosmology),
- SGWB amplitude, tilt, and log-periodic modulation (LISA).

Each observable probes a different aspect of the relaxation dynamics, yet all depend on the same parameter α. This one-parameter structure is a central feature of RDCC and provides a stringent test of the framework.

8.3 Falsifiability

RDCC is sharply falsifiable. Any inconsistency among the correlated predictions would rule out the framework. In particular:

- a measured scalar tilt inconsistent with $n_s - 1 = -2\alpha$,
- a dark-radiation excess not scaling as $\Delta N_{\text{eff}} \sim \alpha$,
- a growth-rate observable incompatible with the predicted percent-level deviation,
- a dark-matter abundance inconsistent with the sectoral ratio ρ_-/ρ_+,
- absence of the predicted SGWB features in the LISA band,

would falsify the RDCC framework.

Conversely, detection of log-periodic oscillations in the SGWB would provide strong evidence for the global CPT structure and the bipartite entanglement across the bounce.

8.4 Summary

The phenomenology of RDCC is governed by a single parameter and exhibits a highly constrained correlation structure. This renders the framework predictive, minimal, and observationally testable across multiple cosmological probes.

9 Conclusion

Relaxation-Driven Cyclic Cosmology (RDCC) provides a minimal and predictive realization of a CPT-symmetric Universe. Its structure is built on three foundational ingredients: a bipartite Hilbert-space factorization, a universal relaxation parameter generated by a Planck-suppressed inter-sector operator, and a cuscuton-induced nonsingular bounce. Together, these components yield a coherent cosmological framework in which the thermodynamic arrow of time, dark-matter abundance, scalar spectral tilt, and gravitational-wave signatures arise from a single underlying principle.

The global CPT postulate implies that the full quantum state of the Universe is time-direction neutral. Sectoral observers experience opposite thermodynamic time orientations due to the asymmetric projection of the global state onto the visible and mirror sectors. This geometric neutrality of the time coordinate provides a natural origin for the sectoral arrows of time without invoking special initial conditions.

The effective field theory of RDCC is deliberately minimal. A single dimension-six operator induces a universal relaxation parameter $\alpha \sim 10^{-2}$, whose renormalization-group flow exhibits an infrared fixed point. The relaxation dynamics govern the rate of entanglement exchange between the sectors and determine correlated deviations from ΛCDM across multiple observables.

The bounce is generated by a cuscuton-like operator that provides controlled violation of the null energy condition without introducing additional propagating degrees of freedom. The resulting modified Friedmann equation yields a smooth, nonsingular transition between contraction and expansion. A phase-space analysis shows that the CPT-symmetric trajectory is a dynamical attractor: antisymmetric perturbations between the sectors are damped as $\alpha(H)$ flows to zero near the bounce.

The perturbation sector inherits its structure from the global CPT symmetry and the relaxation dynamics. RDCC predicts a nearly scale-invariant scalar spectrum with $n_s - 1 = -2\alpha$, a blue-tilted stochastic gravitational-wave background peaked in the millihertz range, and log-periodic oscillations arising from interference between the two CPT-related tensor sectors across the bounce. These oscillations constitute a distinctive observational signature and provide a clear target for LISA.

All deviations from ΛCDM are governed by the single parameter α. This one-parameter structure yields a tightly correlated prediction pattern across early- and late-time cosmology, rendering RDCC sharply falsifiable. Any inconsistency among the correlated predictions would rule out the framework, while detection of the predicted log-periodic tensor modulation would provide strong evidence for the global CPT structure and the bipartite entanglement across the bounce.

RDCC thus offers a structurally complete and observationally testable alternative to standard cosmology, grounded in minimal assumptions and governed by a single universal parameter.

References

[1] R. F. Streater and A. S. Wightman, *PCT, Spin and Statistics, and All That*, Princeton University Press (1964).

[2] S. Weinberg, *The Quantum Theory of Fields, Vol. I*, Cambridge University Press (1995).

[3] N. Afshordi, D. J. H. Chung, M. Doran and G. Geshnizjani, "Cuscuton: A Causal Field Theory with an Infinite Speed of Sound," Phys. Rev. D **75**, 123509 (2007).

[4] Planck Collaboration, "Planck 2018 Results. VI. Cosmological Parameters," Astron. Astrophys. **641**, A6 (2020).

[5] LISA Collaboration, "Laser Interferometer Space Antenna," arXiv:1702.00786 [astro-ph.IM].